

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (Mechanical, AMT)

March-April 2018

ME784/ME734 Computer Aided Engineering

Date: 05.04.2018, Thursday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. Section I and II must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of scientific calculator is allowed.

SECTION- I

- Q1 (a)** Using a simplified description of process of design explain where computational aids might be assistance to the designer. 5
- (b)** Prepare the detailed specifications for a typical CAD workstation with latest hardware & state the different commercially available CAE software. 6

- Q2 (a)** Briefly explain flat panel displays. 3
- (b)** Explain the working principle of plotter. 3
- (c)** Using DDA line algorithm, find the pixel positions between end points (6, 9) and (11, 12). 6

OR

- Q2 (a)** Why can't a vector scan display produce as many colours as a raster scan display? 3
- (b)** Explain interpolated curves and approximated curves. 3
- (c)** Using Bresenham's line algorithm, find the pixel positions between end points (0, 2) and (4, 5). 6

- Q3 (a)** What is transformation? Compare geometric transformation with coordinate transformation. 3
- (b)** Prove that 3D rotations are not commutative. 3
- (c)** A square having end points A(1,1), B(6,1), C(6,6), D(1,6) is rotated by 50° clockwise direction keeping point (6,1) fixed. Find the final coordinates. 6

OR

- Q3 (a)** Explain 2-D transformation for shearing in x – axis & y - axis. 3
- (b)** What are the advantages of homogenous coordinate system? 3
- (c)** Find the reflection of point (5, 5) about a line $y = 3x + 8$. 6

SECTION- II

- Q4 (a)** Compare cubic Bezier curve with Hermite cubic curve. 4
(b) Explain the principle of Rapid Prototyping, its advantages and applications. 3
(c) Explain in brief reverse engineering? What are active scanner and passive scanner? 3

- Q5** Derive weak form and finite element formulation for following governing differential equation using linear element. 7

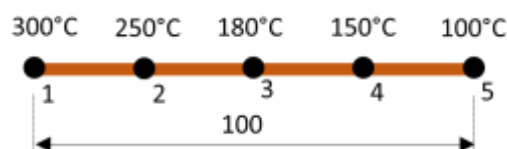
$$C_1 \frac{d^2 u}{dx^2} - C_2 u = 0$$

OR

- Q5** Discuss application of finite element method in solving an engineering problem. 7

- Q6** **Attempt any three:** 18

- (a)** Discuss the advantages, limitations and applications of Finite Difference and Finite Element Methods.
- (b)** Derive the finite difference formula to compute $\frac{dT}{dx}$ using central difference scheme.
- (c)** In a rod having 100 mm length, subjected to heat transfer, equilibrium temperatures at 5 equidistance nodes are shown in figure below. Determine the temperature at point 30 mm and 90 mm from the left wall considering the linear elements.



- (d)** One dimensional deformation inside the one meter long cast steel rod is governed by following equation. Find a three parameter solution using Galerkin Method.

$$\frac{d^2 T}{dx^2} = 2x$$

Subjected to following boundary conditions.

At $x = 0$, $T = 10^\circ\text{C}$; and at $x = 1$, $T = 100^\circ\text{C}$

- (e)** Briefly explain solution algorithm for Jacobi method using an illustration.
